

Mathematics Through Symmetry: Orbit Groupoids, Twin Structures, and a Universal Construction and Classification Theorem

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Abstract

We develop a universal mathematical framework for constructing and classifying structural deformations induced by group symmetries. Given a binary structure $(E, \diamond)$ and a group G acting on E , our procedure canonically generates new operations $\diamond_g$ parameterized by group elements. We prove a universal classification theorem showing that the deformation space is canonically isomorphic to the homogeneous space $L \backslash G$ (right cosets), where L is the structural kernel preserving the original operation. While the individual constructions (e.g. pullback of metrics, conjugation of operators) are classical, our contribution lies in unifying them under a single orbit-theoretic principle. Through purely algebraic and combinatorial examples, we illustrate how this perspective reveals a common mechanism behind apparent diversity in algebra and geometry. The framework does not introduce new physics or solve open problems in analysis; rather, it offers a conceptual lens for classifying symmetry-induced structural change.

1 Introduction

Symmetries play a central role in the foundational organization of mathematics [1, 2]. In category theory, topology, geometry, and algebraic structures, group actions express invariance and govern deformations [4, 5]. In this work, we demonstrate that the essence of structural transformation under symmetry can be captured by a general *transport operation*: a single mechanism that both constructs deformations and classifies them.

Our main contributions are:

- A universal construction generating all group-induced deformations of $(E, \diamond)$;
- A classification theorem proving that all deformations are parametrized by $L \backslash G$, where L is the structural kernel;
- Purely mathematical examples—from deformed group laws to twisted convolutions—illustrating the breadth of the formalism.

This establishes a unified principle applicable across algebra, combinatorics, and geometry.

What this work is *not*

This article does *not*:

- propose a new physical theory unifying gravity and quantum mechanics;
- introduce novel analytical theorems in differential geometry or operator theory;

- claim technical novelty for constructions like metric pullback or operator conjugation.

Rather, it offers a *structural reinterpretation*: it shows that these classical constructions are manifestations of a single algebraic transport mechanism, and that their space of deformations is universally classified by a homogeneous space $L \backslash G$.

Symmetry generates both rigidity and flexibility

Some structures are rigid under symmetries (e.g., classical commutativity, flat geometry), while others exhibit flexibility (curvature, noncommutativity). We show that both phenomena result from whether $G = L$ or $[G : L] > 1$.

Construct *and* classify

A key aspect of our contribution is that the framework:

- constructs all possible symmetry-driven deformations; and
- classifies them completely using the single invariant $L \backslash G$.

No additional data is needed: the action of G on $(E, \diamond)$ determines the entire space of outcomes.

2 Orbit Classification by Group Actions

A group action $G \curvearrowright E$ induces an equivalence relation: $x \sim y \iff \exists g \in G : y = g \cdot x$. The equivalence classes $G \cdot x$ are the *orbits* [3].

Theorem 2.1 (Orbit Partition Theorem). *E decomposes into disjoint orbits:*

$$E = \bigsqcup_{x \in E/G} G \cdot x.$$

Definition 2.2 (Stabilizer). $G_x := \{g \in G : g \cdot x = x\}$.

Theorem 2.3 (Orbit–Stabilizer Theorem). *There is a G -equivariant bijection*

$$G/G_x \simeq G \cdot x.$$

Remark 2.4. *Homogeneous spaces G/G_x classify configurations of x generated by symmetries.*

3 Preliminaries

Let G be a group acting on a nonempty set E by $G \times E \rightarrow E$, $(g, x) \mapsto g \cdot x$. We fix throughout a binary operation $\diamond : E \times E \rightarrow E$.

Transport of the structure by $g \in G$ is defined by

$$x \diamond_g y := g^{-1} \cdot ((g \cdot x) \diamond (g \cdot y)), \tag{1}$$

which yields a new binary structure $(E, \diamond_g)$.

Remark 3.1. *This construction generalizes conjugation in groups and twisted products in non-commutative geometry [6].*

Denote the set of transported operations by $\mathcal{O} := \{\diamond_g : g \in G\}$.

Box 1: Operations vs. Objects – An Inversion of Priority

Traditionally, one transports *objects* (functions, tensors, operators) under group actions, keeping the algebraic structure fixed. Here, we fix the underlying set E and *transport the operation itself*. This inversion reveals that what varies is not the carrier set, but the *rule of composition*. The formalism $x \diamond_g y := g^{-1}((g \cdot x) \diamond (g \cdot y))$ makes this inversion explicit and universal.

Concrete illustration: On $\mathbb{R}$ with addition, if $g : x \mapsto x + c$, the transported operation becomes $x \oplus_g y = x + y + c$. The carrier set $\mathbb{R}$ is unchanged; only the rule of combination has shifted.

4 Twin Spaces

We call

$$g \cdot (E, \diamond) := (E, \diamond_g)$$

a *twin space*. Thus G acts on the set of all twin spaces, forming a groupoid.

More precisely, the relevant object is the *transport (action) groupoid* $G \ltimes \mathcal{O}$: its objects are the twin spaces $\diamond_g \in \mathcal{O}$, and a morphism $\diamond_g \rightarrow \diamond_{g'}$ is an element $h \in G$ carrying the first to the second. Since every twin space is reached from $\diamond$, this groupoid is connected, and its isotropy (vertex) group at the base point $\diamond$ is exactly the structural kernel L . A connected groupoid is equivalent to its vertex group, so $G \ltimes \mathcal{O} \simeq L$ as a groupoid, while its set of objects is the homogeneous space $\mathcal{O} \cong L \backslash G$ of the classification theorem below. The groupoid therefore packages both halves of the theory at once: its objects record the *deformations* $L \backslash G$, and its vertex group records the *symmetries* L .

Lemma 4.1. *The assignment $g \mapsto g \cdot (E, \diamond)$ defines a transitive right action of G on the set of twin spaces (transport composes on the right: $(\diamond_g)_h = \diamond_{gh}$).*

Proof. Immediate from (1). □

Twin structures inherit many algebraic and geometric properties of the original structure, but often with transformed parameters.

5 Twin Operations and the Structural Kernel

We now introduce the map that associates to each $g \in G$ its transported operation $\diamond_g$. This defines the *deformation morphism* $\pi : G \rightarrow \mathcal{O}$, $\pi(g) = \diamond_g$.

Definition 5.1 (Structural kernel). *The structural kernel is*

$$L := \{g \in G : \diamond_g = \diamond\} = \{g \in G : g \cdot (x \diamond y) = (g \cdot x) \diamond (g \cdot y)\}.$$

Lemma 5.2. *L is a subgroup of G .*

Proof. The map $g \mapsto \diamond_g$ is an action of G on $\mathcal{O}$ and L is the stabiliser of the base point $\diamond$, so it is a subgroup. Explicitly $\diamond_e = \diamond$ gives $e \in L$; if $\diamond_g = \diamond$ then applying the transport by g^{-1} gives $\diamond_{g^{-1}} = \diamond$; and if $\diamond_g = \diamond_h = \diamond$ then $\diamond_{gh} = \diamond$ by the composition law of (1). □

Remark 5.3. *Elements of L preserve the original structure. Thus L encodes the internal symmetry of $(E, \diamond)$.*

6 Universal Construction and Classification Theorem

We now state our main result, showing that the transported structures form a homogeneous space.

Theorem 6.1 (Universal Construction and Classification Theorem). *There is a canonical bijection $\mathcal{O} \simeq L \backslash G$, given by $Lg \mapsto \diamond_g$. The fibers of π are the right cosets Lg ; thus $\mathcal{O} \cong L \backslash G$ as a homogeneous space for the right transport action.*

Proof. Surjectivity of π follows by definition of $\mathcal{O}$.

Let $g_1, g_2 \in G$. Then $\diamond_{g_1} = \diamond_{g_2}$ iff

$$g_2 g_1^{-1} \cdot (x \diamond y) = (g_2 g_1^{-1} \cdot x) \diamond (g_2 g_1^{-1} \cdot y) \quad \forall x, y \in E,$$

i.e. iff $g_2 g_1^{-1} \in L$. Thus $\diamond_{g_1} = \diamond_{g_2}$ exactly when g_1, g_2 lie in the same right coset (indeed $g_2 g_1^{-1} \in L \iff Lg_1 = Lg_2$).

The fibers of π are thus precisely the right cosets Lg . Hence $\mathcal{O} \simeq L \backslash G$. $\square$

Corollary 6.2 (Universality). *Every transported structure in $\mathcal{O}$ arises uniquely from some $g \in G$, modulo the structural kernel L :*

$$\mathcal{O} = \{\diamond_g : g \in G\} \quad \text{and} \quad \mathcal{O} \simeq L \backslash G.$$

No additional data is needed to generate or parametrize deformations.

Corollary 6.3 (Rigidity versus flexibility).

$$G = L \iff \mathcal{O} = \{\diamond\}.$$

Nontrivial transported structures exist iff $[G : L] > 1$.

Remark 6.4. *The deformation space depends only on the action of G and not on the specific form of $\diamond$ beyond its symmetry group.*

Remark 6.5 (Right cosets, and normality). *The fibers of π are the right cosets Lg (the condition is $g_2 g_1^{-1} \in L$); the canonical model is therefore $L \backslash G$, not G/L . Inversion $g \mapsto g^{-1}$ is a bijection $L \backslash G \rightarrow G/L$, so the number of deformations is the index $[G : L]$ either way, but the natural assignment $g \mapsto \diamond_g$ is constant precisely on right cosets. The two coset spaces coincide canonically iff $L \trianglelefteq G$. In the affine examples below, L is not normal, and the deformation parameter is exactly a right-coset invariant.*

7 Purely Mathematical Examples

We now illustrate the universality of the transport mechanism through *purely mathematical* (non-physical) examples.

Example 7.1 (Translated addition on $\mathbb{Z}$). *Let $E = \mathbb{Z}$ with $\diamond = +$. Let $G = \text{Aff}(\mathbb{Z}) = \{x \mapsto \varepsilon x + b \mid \varepsilon = \pm 1, b \in \mathbb{Z}\}$. For $g(x) = x + 1$, we obtain*

$$x \oplus_g y = g^{-1}(g(x) + g(y)) = ((x + 1) + (y + 1)) - 1 = x + y + 1.$$

This operation remains associative: $(x \oplus_g y) \oplus_g z = x + y + z + 2 = x \oplus_g (y \oplus_g z)$. In fact, $(E, \oplus_g)$ is isomorphic to $(E, +)$ via the translation $x \mapsto x + 1$.

Here $L = \{x \mapsto \varepsilon x \mid \varepsilon = \pm 1\}$, and $L \backslash G \cong \mathbb{Z}$ parametrizes a family of “translated” additions, all mutually isomorphic but formally distinct as operations on $\mathbb{Z}$.

Example 7.2 (Twisted convolution algebras). Let $E = L^1(G)$ with convolution $*$. For $\alpha \in \text{Aut}(G)$, define

$$f *_\alpha h := \alpha^{-1}(\alpha(f) * \alpha(h)).$$

Because convolution is equivariant under every automorphism, $\alpha(f) * \alpha(h) = \alpha(f * h)$ (up to the modular factor on non-unimodular G), so $f *_\alpha h = f * h$ for all α . Hence the structural kernel is $L = \text{Aut}(G)$ and the deformation is trivial: convolution is rigid under automorphism transport. Genuinely “twisted” convolution algebras instead arise from a 2-cocycle $\sigma \in Z^2(G, \mathbb{T})$ (the twisted group algebra $\mathbb{C}_\sigma[G]$) — a cohomological deformation, not a transport, and so outside the present mechanism.

Example 7.3 (Deformed lattice operations). Let $E = \mathcal{P}(X)$ with intersection $\cap$. Let $G = \text{Sym}(X)$ act by permutation: $\sigma \cdot A = \sigma(A)$. Then $A \cap_\sigma B = \sigma^{-1}(\sigma(A) \cap \sigma(B)) = A \cap B$, so $L = G$: the lattice is rigid. This illustrates that set-theoretic operations often exhibit maximal rigidity under natural symmetry groups.

8 Four Explicit Computations of $L \backslash G$

We now present detailed computations of the deformation space for various structures.

8.1 Example 1: Deformed addition on $\mathbb{R}$ under affine transformations

Setup:

- Set: $E = \mathbb{R}$
- Operation: $x \diamond y = x + y$
- Group: $G = \text{Aff}^+(\mathbb{R}) = \{g_{a,b} : x \mapsto ax + b \mid a > 0, b \in \mathbb{R}\}$

Computation: For $g = g_{a,b}$, we have $g^{-1}(x) = (x - b)/a$, so

$$x \diamond_g y = g^{-1}(g(x) + g(y)) = \frac{(ax + b) + (ay + b) - b}{a} = x + y + \frac{b}{a}.$$

Structural kernel: $\diamond_g = \diamond$ iff $b/a = 0$ iff $b = 0$. Thus

$$L = \{x \mapsto ax \mid a > 0\} \cong (\mathbb{R}_{>0}, \times).$$

Classification: $L \backslash G \cong \mathbb{R}$, with the right coset $Lg_{a,b}$ determined by $b/a \in \mathbb{R}$. Note L is *not* normal in G , so $L \backslash G \neq G/L$ canonically; b/a is the right-coset invariant ($s_\alpha g_{a,b} = g_{\alpha a, \alpha b}$ fixes it), whereas the left coset $g_{a,b}L$ is labelled by b .

The transported operations $x \oplus_c y = x + y + c$ for $c \in \mathbb{R}$ form a one-parameter family of group laws on $\mathbb{R}$, all isomorphic to $(\mathbb{R}, +)$.

8.2 Example 2: Conjugation action on matrix multiplication (Rigidity)

Setup:

- Set: $E = M_n(\mathbb{C})$
- Operation: $A \diamond B = AB$ (matrix multiplication)
- Group: $G = \text{GL}_n(\mathbb{C})$, acting by conjugation: $g \cdot A = gAg^{-1}$

Computation:

$$\begin{aligned} A \diamond_g B &= g^{-1} \cdot ((g \cdot A) \diamond (g \cdot B)) \\ &= g^{-1} \cdot ((gAg^{-1})(gBg^{-1})) \\ &= g^{-1} \cdot (gABg^{-1}) \\ &= g^{-1}(gABg^{-1})g = AB = A \diamond B. \end{aligned}$$

Structural kernel: $L = G$ (every element preserves the operation).

Classification: $L \backslash G = \{*\}$ (single point). The deformation space is trivial: $\mathcal{O} = \{\diamond\}$.

Interpretation: Matrix multiplication exhibits *complete structural rigidity* under conjugation. This reflects the fact that conjugation is an inner automorphism of the algebra $M_n(\mathbb{C})$.

8.3 Example 3: Cyclic group acting on a finite magma

Setup:

- Set: $E = \{0, 1, 2\}$
- Operation: Define the magma $(E, \diamond)$ by the following Cayley table:

$\diamond$	0	1	2
0	0	1	2
1	2	0	1
2	1	2	0

- Group: $G = C_3 = \langle \sigma \rangle$ where $\sigma = (0 \ 1 \ 2)$ acts by cyclic permutation.

Computation: The transported table under σ is:

$$x \diamond_{\sigma} y = \sigma^{-1}(\sigma(x) \diamond \sigma(y)).$$

For instance, $0 \diamond_{\sigma} 1 = \sigma^{-1}(1 \diamond 2) = \sigma^{-1}(1) = 0$.

Computing the full table:

$\diamond_{\sigma}$	0	1	2
0	2	0	1
1	1	2	0
2	0	1	2

This is distinct from $(E, \diamond)$ since $1 \diamond 0 = 2 \neq 1 = 1 \diamond_{\sigma} 0$.

Structural kernel: Direct verification shows $L = \{e\}$ (only the identity preserves the table).

Classification: $L \backslash G \cong C_3$, yielding three distinct operations $\diamond, \diamond_{\sigma}, \diamond_{\sigma^2}$.

8.4 Example 4: Two actions on $(\mathbb{R}_{>0}, \cdot)$

This example shows how the *same operation* can exhibit different rigidity depending on the group action.

Setup: $E = \mathbb{R}_{>0}$, $\diamond = \cdot$ (multiplication).

Action 1: Multiplicative scaling. Let $G = (\mathbb{R}, +)$ act by $t \cdot x = e^t x$.

Then $x \diamond_t y = e^{-t}(e^t x \cdot e^t y) = e^t xy \neq xy$ for $t \neq 0$.

Structural kernel: $L = \{0\}$. Quotient: $L \backslash G = \mathbb{R}$ (infinite deformation space).

Action 2: Power action. Let $G = (\mathbb{R}_{>0}, \times)$ act by $a \cdot x = x^a$.

Then $x \diamond_a y = (x^a \cdot y^a)^{1/a} = (xy)^{a \cdot (1/a)} = xy$.

Structural kernel: $L = G$. Quotient: $L \backslash G = \{*\}$ (complete rigidity).

Conclusion: The deformation space depends crucially on *how* the group acts, not just on the group or the operation individually.

9 Conceptual Contribution

While the orbit-stabilizer theorem is classical, its systematic application to *operation deformation*—rather than point orbits—appears underexploited in the literature. Our framework makes this application explicit and uniform.

A unified viewpoint

The transport formula $\diamond_g = g^{-1} \circ \diamond \circ (g \times g)$ reveals that:

- Metric pullback under diffeomorphisms (geometry),
- Product deformation under automorphisms (algebra),
- Conjugation of group laws (group theory)

are instances of a *single mechanism*. This is not merely a metaphor but a precise structural identity.

Classification without cohomology

In some contexts, deformation classification requires cohomological machinery (e.g., $H^2(G, A)$ for group extensions). The present framework provides an alternative when deformations arise purely from transport: the classification is immediate from $L \backslash G$ without spectral sequences or cocycle computations. The two viewpoints are complementary rather than opposed: the global orbit $L \backslash G$ is the set of finite deformations of transport type, and its infinitesimal structure at the base point—the tangent space $\mathfrak{g}/\mathfrak{l}$ of the homogeneous space—sits inside the first-order deformation space $H^1(G, \text{End } E)$ classified cohomologically in the companion deformation paper. Transport deformations are the integrable, transport-type directions; genuinely cohomological deformations (cocycles that are not transports, such as the 2-cocycle twists above) lie in the complement. Conversely, genuinely cohomological deformations — such as 2-cocycle (projective) twists, as in the convolution example above — are *not* of transport type and lie outside this framework.

Distinguishing rigidity from flexibility

The dichotomy $G = L$ (rigidity) versus $[G : L] > 1$ (flexibility) provides a simple criterion for determining whether a structure admits nontrivial deformations under a given symmetry group.

10 Categorical Perspective

Proposition 10.1 (Universal property). *Let $\mathcal{C}$ be a category with a forgetful functor to **Set**. Let $(E, \diamond) \in \mathcal{C}$, and let G act on the underlying set. Then the transport $\diamond_g$ is the unique operation making the diagram*

$$\begin{array}{ccc} E \times E & \xrightarrow{\diamond} & E \\ g^{-1} \times g^{-1} \downarrow & & \downarrow g^{-1} \\ E \times E & \xrightarrow{\diamond_g} & E \end{array}$$

*commute in **Set**. Thus $\diamond_g = g^{-1} \circ \diamond \circ (g \times g)$ is the canonical equivariant lift.*

Proof. Commutativity of the square reads $g^{-1} \circ \diamond = \diamond_g \circ (g^{-1} \times g^{-1})$ as maps $E \times E \rightarrow E$. Precomposing with the bijection $g \times g$ gives $g^{-1} \circ \diamond \circ (g \times g) = \diamond_g$, which both establishes existence and determines $\diamond_g$ uniquely, since g^{-1} and $g \times g$ are invertible. This is precisely formula (1). $\square$

This shows the transport is not ad hoc, but the unique operation compatible with equivariance.

11 Conclusion

We introduced the theory of twin structures: transported operations arising from group symmetries. The following key outcomes were established:

1. A universal method to generate all group-induced deformations of a binary structure $(E, \diamond)$;
2. A classification theorem proving the canonical identification $\mathcal{O} \simeq L \backslash G$, where L is the structural kernel;
3. Explicit computations demonstrating the framework across algebra, combinatorics, and analysis.

This perspective reveals that seemingly disparate phenomena—translated group laws, twisted convolutions, rigid lattices—share a common structural origin. The framework provides a *conceptual unifier* for symmetry-induced structural change.

Future Directions

- Classification of transported n -ary structures;
- Topological and metric analysis on the homogeneous space $L \backslash G$;
- Connections with formal deformation theory and Hochschild cohomology;
- Applications to formal language theory and combinatorial algebra.

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